

ATOMS

Important formulas and concepts directly from the NCERT-

- First model of atom was **plum pudding model** given by **J.J. Thomson** acc to which the positive charge is uniformly distributed throughout the volume of atom and the negatively charged electrons are embedded in it like seeds in watermelon.
- According to **rutherford's nuclear model of atom**, the entire positive charge and most of the mass of atom is concentrated in small volume called the nucleus. with electrons revolving around the nucleus.

❖ Alpha-Particle and Rutherford's Scattering model of atom-

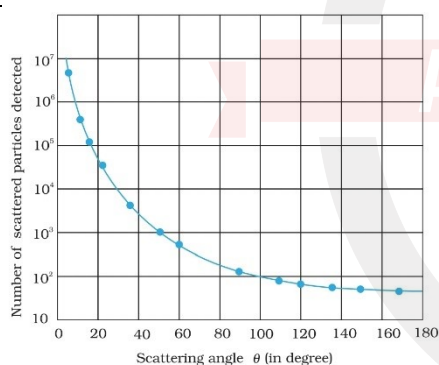


FIGURE 12.3 The dots are α -particle scattering data for a thin gold foil obtained by Geiger and Marsden using the experiment shown in Figs. 12.1 and 12.2. The solid curve is the theoretical prediction based on the assumption that atom has a small, dense, positively charged nucleus.

- Acc to this model, the size of atom is 10,000 to 100,000 times larger than the nucleus.
- Most of atom is empty space due to which majority alpha particles pass undeflected and alpha particles which pass near nucleus, deflects through some angles.
- **The atomic electrons being so light do not appreciably affect the alpha particles.**
- The force b/w alpha Particle and nucleus,

$$F = \frac{1}{4\pi\epsilon_0} \left(\frac{2e \cdot Ze}{r^2} \right)$$

Where, r is the distance b/w alpha particle and nucleus.

○ **Alpha particle trajectory-**

- **Impact parameter** is the perpendicular distance of the initial velocity vector of α -particle from the centre of the nucleus.
- In case of head on collision, the impact parameter is minimum and α -particle rebounds back.
- For a large impact parameter, the α -particle goes nearly undeviated and has a small deflection.

○ **Electron orbits-**

- Relation b/w orbit radius and electron velocity,

$$r = e^2 / 4\pi\epsilon_0 m v^2$$

The kinetic energy (K) and electrostatic potential energy (U) of the electron in hydrogen atom are

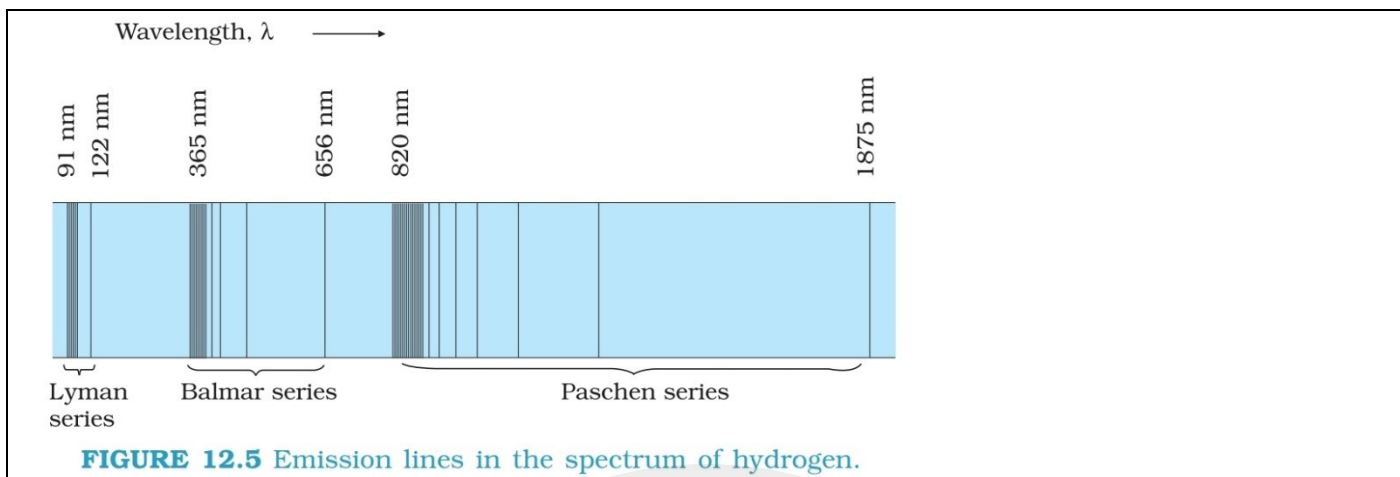
$$K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r} \text{ and } U = -\frac{e^2}{4\pi\epsilon_0 r}$$

(The negative sign in U signifies that the electrostatic force is in the $-r$ direction.) Thus the total energy E of the electron in a hydrogen atom is

$$\begin{aligned} E = K + U &= \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} \\ &= -\frac{e^2}{8\pi\epsilon_0 r} \end{aligned}$$

The total energy of the electron is negative. This implies the fact that the electron is bound to the nucleus. If E were positive, an electron will not follow a closed orbit around the nucleus.

❖ **Atomic spectra-**



❖ Bohr model of hydrogen atom-

- Acc. To electromagnetic theory, an accelerated electron must spiral into nucleus as it loses energy in form of radiation.
- Postulates of bohr model-
 - 1) An electron in an atom could revolve in certain stable orbits without emission of energy. These orbits have definite total energy, k/as stationary states of the atom.
 - 2) Electron revolves only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$. Thus the angular momentum of orbital electron is quantised.

$$L = nh/2\pi$$

- 3) If an electron makes a transition from orbit of higher energy to lower energy then a photon is emitted having energy equal to energy diff b/w initial and final states. The energy of emitted photon,

$$h\nu = E_i - E_f$$

- Radius of nth possible orbit,

$$r = (n^2/m)(h^2/2\pi) 4\pi\epsilon_0/e^2$$

- Total energy of electron in stationary orbits,

$$E = - (me^4/8n^2\epsilon_0^2h^2)$$

Or in eV,

$$E = (-13.6/n^2) \text{ eV}$$

Or in Joule,

$$E = (-2.18 \times 10^{-18}/n^2) \text{ J}$$

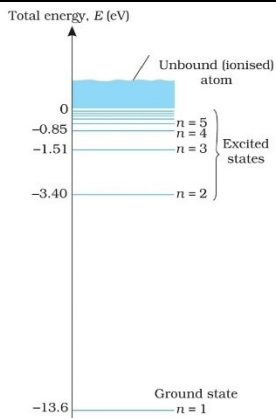


FIGURE 12.8 The energy level diagram for the hydrogen atom. The electron in a hydrogen atom at room temperature spends most of its time in the ground state. To ionise a hydrogen atom an electron from the ground state, 13.6 eV of energy must be supplied. (The horizontal lines specify the presence of allowed energy states.)

❖ The line spectra of the Hydrogen atom-k

$$h\nu_{if} = E_{n_i} - E_{n_f}$$

$$h\nu_{if} = \frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (12.21)$$

$$\text{or } \nu_{if} = \frac{me^4}{8\epsilon_0^2 h^3} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Equation (12.21) is the Rydberg formula, for the spectrum of the hydrogen atom. In this relation,

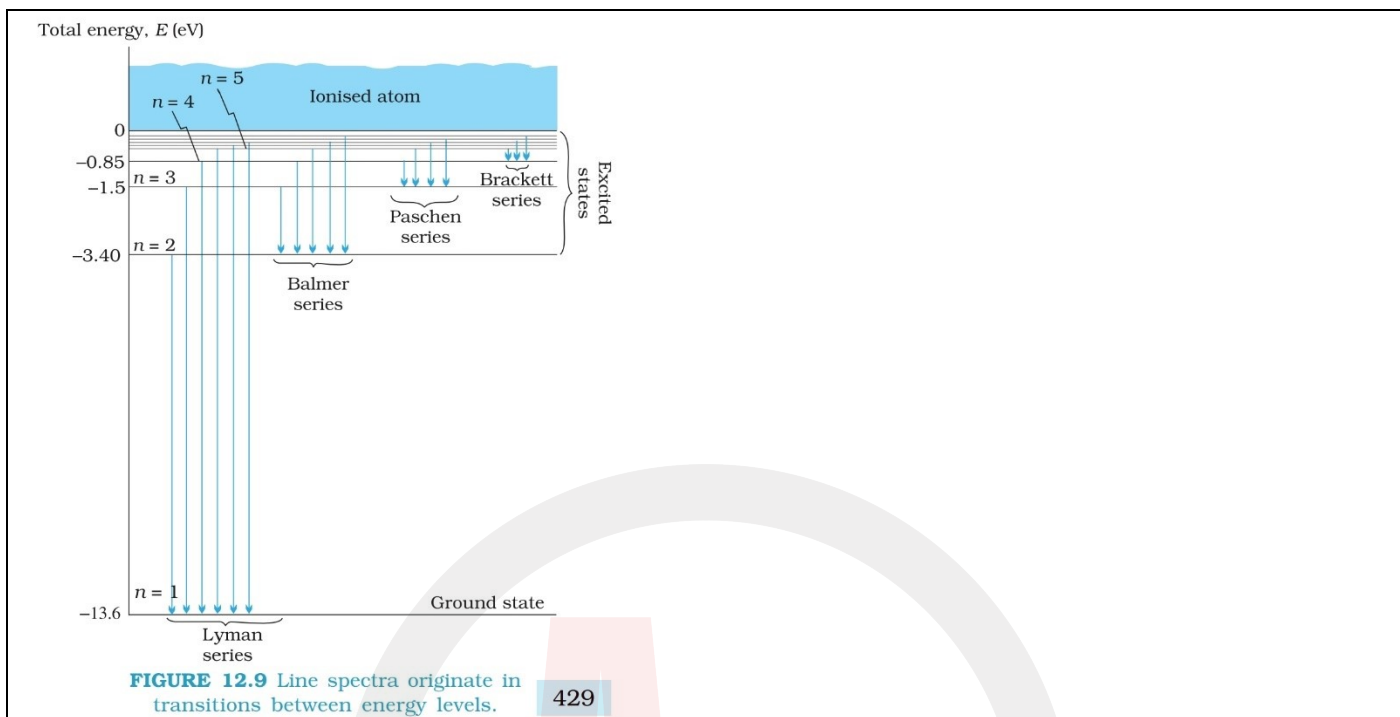
The Rydberg constant R is readily identified to be

$$R = \frac{me^4}{8\epsilon_0^2 h^3 c} \quad (12.23)$$

If we insert the values of various constants in Eq. (12.23), we get

$$R = 1.03 \times 10^7 \text{ m}^{-1}$$

- When electrons jump from higher energy states to a common lower energy state, they form a series like Lyman, Balmer, Paschen, Bracket and Pfund series.



- The various lines are produced when electrons jump from higher to lower energy state and photons are emitted. These spectral lines are called **emission lines**. but when an atom absorbs a photon, process is k/as absorption.

❖ De Broglie's explanation of Bohr's Second Postulate of Quantisation-

- For an electron moving in n^{th} circular orbit of radius r_n , the total distance is the circumference of the orbit, thus

$$2\pi r_n = n\lambda, \quad n = 1, 2, 3, \dots$$

- The quantised electron orbits and energy states are due to the wave nature of electron.

❖ Limitations of Bohr atomic model-

- 1) It is applicable to hydrogenic atoms (single electronic atoms).
 - 2) It is unable to explain the relative intensities of the frequencies in the spectrum. This model is unable to account for the intensity variations.
- Bohr's model can't be used for complex atoms, for them we have to use **Quantum Mechanics**.

SUMMARY

1. Atom, as a whole, is electrically neutral and therefore contains equal amount of positive and negative charges.
2. In *Thomson's model*, an atom is a spherical cloud of positive charges with electrons embedded in it.
3. In *Rutherford's model*, most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus (typically one by ten thousand the size of an atom), and the electrons revolve around it.
4. Rutherford nuclear model has two main difficulties in explaining the structure of atom: (a) It predicts that atoms are unstable because the accelerated electrons revolving around the nucleus must spiral into the nucleus. This contradicts the stability of matter. (b) It cannot explain the characteristic line spectra of atoms of different elements.
5. Atoms of most of the elements are stable and emit characteristic spectrum. The spectrum consists of a set of isolated parallel lines termed as line spectrum. It provides useful information about the atomic structure.
6. To explain the line spectra emitted by atoms, as well as the stability of atoms, Niel's Bohr proposed a model for hydrogenic (single electron) atoms. He introduced three postulates and laid the foundations of quantum mechanics:
 - (a) In a hydrogen atom, an electron revolves in certain stable orbits (called stationary orbits) without the emission of radiant energy.
 - (b) The stationary orbits are those for which the angular momentum is some integral multiple of $h/2\pi$. (Bohr's quantisation condition.) That is $L = nh/2\pi$, where n is an integer called the principal quantum number.

- (c) The third postulate states that an electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states. The frequency (ν) of the emitted photon is then given by

$$h\nu = E_i - E_f$$

An atom absorbs radiation of the same frequency the atom emits, in which case the electron is transferred to an orbit with a higher value of n .

$$E_i + h\nu = E_f$$

7. As a result of the quantisation condition of angular momentum, the electron orbits the nucleus at only specific radii. For a hydrogen atom it is given by

$$r_n = \left(\frac{n^2}{m} \right) \left(\frac{h}{2\pi} \right)^2 \frac{4\pi\epsilon_0}{e^2}$$

The total energy is also quantised:

$$E_n = -\frac{me^4}{8n^2\epsilon_0^2h^2}$$
$$= -13.6 \text{ eV}/n^2$$

The $n = 1$ state is called ground state. In hydrogen atom the ground state energy is -13.6 eV . Higher values of n correspond to excited states ($n > 1$). Atoms are excited to these higher states by collisions with other atoms or electrons or by absorption of a photon of right frequency.

8. de Broglie's hypothesis that electrons have a wavelength $\lambda = h/mv$ gave an explanation for Bohr's quantised orbits by bringing in the wave-particle duality. The orbits correspond to circular standing waves in which the circumference of the orbit equals a whole number of wavelengths.
9. Bohr's model is applicable only to hydrogenic (single electron) atoms. It cannot be extended to even two electron atoms such as helium. This model is also unable to explain the relative intensities of the frequencies emitted even by hydrogenic atoms.

POINTS TO PONDER

1. Both the Thomson's as well as the Rutherford's models constitute an unstable system. Thomson's model is unstable electrostatically, while Rutherford's model is unstable because of electromagnetic radiation of orbiting electrons.
2. What made Bohr quantise angular momentum (second postulate) and not some other quantity? Note, h has dimensions of angular momentum, and for circular orbits, angular momentum is a very relevant quantity. The second postulate is then so natural!
3. The orbital picture in Bohr's model of the hydrogen atom was inconsistent with the uncertainty principle. It was replaced by modern quantum mechanics in which Bohr's orbits are regions where the electron may be found with large probability.
4. Unlike the situation in the solar system, where planet-planet gravitational forces are very small as compared to the gravitational force of the sun on each planet (because the mass of the sun is so much greater than the mass of any of the planets), the electron-electron electric force interaction is comparable in magnitude to the electron-nucleus electrical force, because the charges and distances are of the same order of magnitude. This is the reason why the Bohr's model with its planet-like electron is not applicable to many electron atoms.
5. Bohr laid the foundation of the quantum theory by postulating specific orbits in which electrons do not radiate. Bohr's model include only one quantum number n . The new theory called quantum mechanics supportes Bohr's postulate. However in quantum mechanics (more generally accepted), a given energy level may not correspond to just one quantum state. For example, a state is characterised by four quantum numbers (n , l , m , and s), but for a pure Coulomb potential (as in hydrogen atom) the energy depends only on n .
6. In Bohr model, contrary to ordinary classical expectation, the frequency of revolution of an electron in its orbit is not connected to the frequency of spectral line. The later is the difference between two orbital energies divided by h . For transitions between large quantum numbers (n to $n - 1$, n very large), however, the two coincide as expected.
7. Bohr's semiclassical model based on some aspects of classical physics and some aspects of modern physics also does not provide a true picture of the simplest hydrogenic atoms. The true picture is quantum mechanical affair which differs from Bohr model in a number of fundamental ways. But then if the Bohr model is not strictly correct, why do we bother about it? The reasons which make Bohr's model still useful are:

- (i) The model is based on just three postulates but accounts for almost all the general features of the hydrogen spectrum.
- (ii) The model incorporates many of the concepts we have learnt in classical physics.
- (iii) The model demonstrates how a theoretical physicist occasionally must quite literally ignore certain problems of approach in hopes of being able to make some predictions. If the predictions of the theory or model agree with experiment, a theoretician then must somehow hope to explain away or rationalise the problems that were ignored along the way.



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